Roll No. ....

TEE–305

B. TECH. (ELECTRICAL ENGINEERING)
(THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

ANALOG ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Describe the construction and working of a PN junction diode. Also draw the static VI characteristics of the diode. (CO1)

OR

(b) What is Varactor diode ? Explain its working. (CO1)

2. (a) Explain the working of a Zener diode as a voltage regulator. (CO2)

OR

(b) Explain the working of a diode based full wave bridge rectifier. Also, draw its complete circuit diagram and input/output waveforms. (CO2)

3. (a) Explain the working of a BJT in saturation and cut-off region. (CO2)

OR

(b) What are the various biasing of BJT ? Explain any *one* in detail. (CO2)

P. T. O.

4. (a) Discuss the input and output characteristics of BJT in common emitter configuration. (CO3)

OR

(b) For a BJT with emitter biasing, R_B = 480 kΩ, R_C = 3 kΩ and R_E = 2 kΩ, V_{CC} = + 20 V and C_E = 50 μF. Calculate the value of base current, collector current, emitter to collector voltage and collector voltage if the value of common emitter current amplification factor is 100. (CO3)

5. (a) Explain the construction and working of Enhancement type MOSFET. (CO2)

OR

(b) Determine the value of drain current of enhancement MOSFET if $ID_{(ON)}$ = 20 mA; $V_{GS(ON)}$ = 8 V; V_T = 2 V and V_{GS} = 4 V. (CO3)